

$$|e_{n+1}| \leq (1+kh)|e_n| + mh^2$$

$$\frac{|e_{n+1}|}{(1+kh)^{n+1}} = |U_{n+1}| \leq \dots$$

$$|U_{n+1}| \leq \frac{|e_n|}{(1+kh)^n} + \frac{mh^2}{(1+kh)^{n+1}}$$

$$\leq |U_n| + \frac{mh^2}{(1+kh)^{n+1}}$$

$$|U_{n+1}| - |U_n| \leq \frac{mh^2}{(1+kh)^{n+1}}$$

$$\sum_{i=0}^{n-1} (|U_{i+1}| - |U_i|) \leq \sum_{i=0}^{n-1} \frac{mh^2}{(1+kh)^{i+1}} = \frac{mh^2}{(1+kh)} \sum_{i=0}^{n-1} \frac{1}{(1+kh)^i}$$

$$= \frac{mh^2}{1+kh} \times \frac{1 - \left(\frac{1}{1+kh}\right)^n}{1 - \frac{1}{1+kh}}$$

$$= \frac{mh^2}{1+kh} \times \frac{1 - (\dots)^n}{\frac{kh}{1+kh}}$$

$$= \frac{mh}{k} \left(1 - \left(\frac{1}{1+kh}\right)^n\right) \leq \frac{mh}{k}$$

$$3. \quad \sum_{i=0}^{n-1} (|U_{i+1}| - |U_i|) = |U_n| - |U_0| \leq \frac{mh}{k}$$

$$|U_n| \leq |U_0| + \frac{mh}{k}$$

$$\left| \frac{e_n}{(1+kh)^n} \right| \leq |e_0| + \frac{mh}{k}$$

$$|e_n| \leq (1+kh)^n \left(|e_0| + \frac{mh}{k} \right)$$

$$4. \begin{cases} (e^x - (1+x))' = e^x - 1 \\ (e^x - 1)' = e^x \end{cases}$$

$$\begin{cases} e^0 = 1 > 0 \end{cases}$$

$$\begin{cases} e^0 - 1 = 0 \Rightarrow e^x - 1 > 0 \text{ pour } x > 0 \\ e^0 - (1+0) = 0 \end{cases}$$

Donc pour $x > 0$ $e^x - (1+x) \geq 0$

$$(1+x) \leq e^x \Leftrightarrow (1+kh) \leq e^{kh} \text{ pour } h > 0$$

$$|e_n| \leq (1+kh)^n \left(|e_0| + \frac{mh}{k} \right)$$

$$|e_n| \leq e^{nkh} \left(\underline{|e_0|} + \frac{mh}{k} \right)$$

5. quand $\tilde{y}_0 = y(t_0)$ on a $|e_0| = 0$

$$|e_n| \leq e^{nkh} \times \frac{mh}{k}$$

On a $t_n = t_0 + nh$ fixé, alors $h \rightarrow 0$ et $n \rightarrow \infty$
 $nh = t - t_0$ reste constante.

Donc $khn = k(t - t_0)$ constante.

$$\lim_{h \rightarrow 0} |e_n| \leq \lim_{h \rightarrow 0} e^{k(t-t_0)} \times \frac{mh}{k} = e^{k(t-t_0)} \times \frac{m \times 0}{k} = 0$$

Alors